

Mathematics

A Cultural Heritage

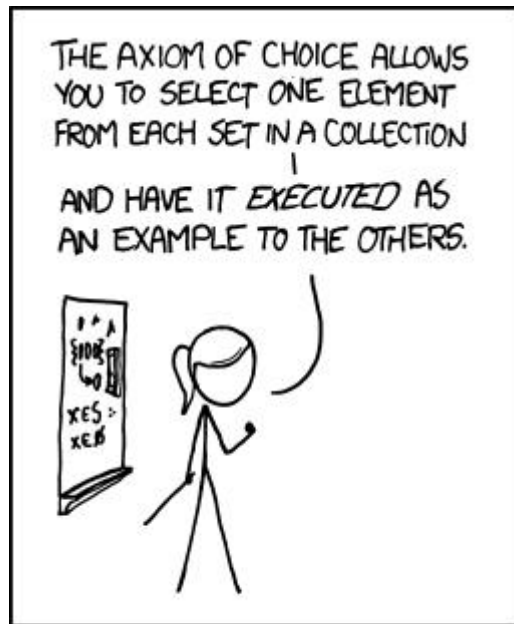
Lecture 1

Introduction: Theorems and Proofs

Course Topics:

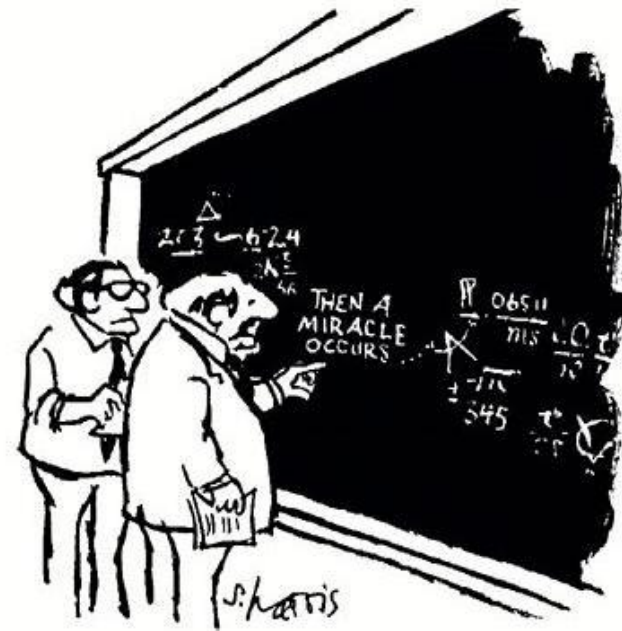
Theorems and Proofs

Mathematical Theorems and their Proofs



MY MATH TEACHER WAS A BIG BELIEVER IN PROOF BY INTIMIDATION.

Source: xkcd



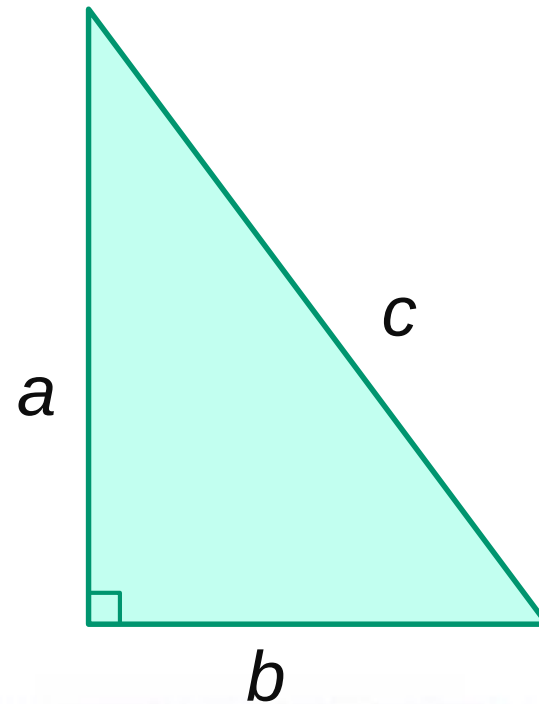
"I THINK YOU SHOULD BE MORE EXPLICIT HERE IN STEP TWO."

From: What's so funny about science
By: Sidney Harris

The Pythagorean Theorem

Theorem In any right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$a^2 + b^2 = c^2$$



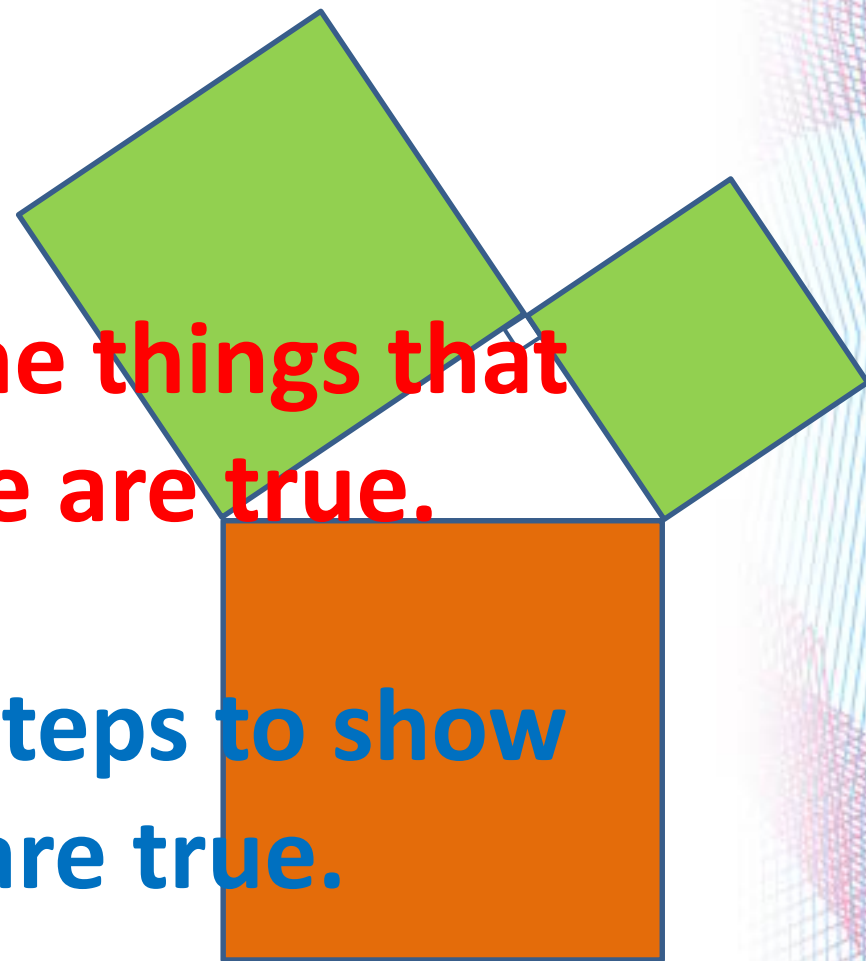
Euclid's *Elements*, Book I, Proposition 47

Proposition In right-angled triangles the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.

Theorems are the things that we can prove are true.

Proofs are the steps to show that they are true.

(Here the squares are interpreted as areas)



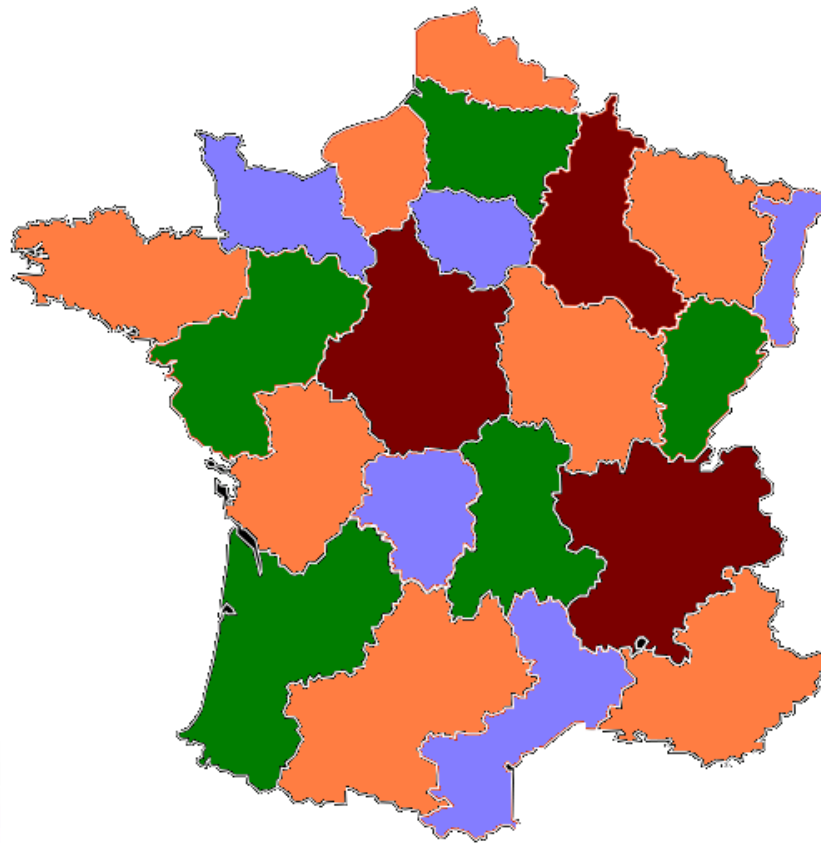
Irrationality of $\sqrt{2}$

Theorem There **do not** exist integers m and n such that $\sqrt{2} = \frac{m}{n}$. In other words, $\sqrt{2}$ is irrational.

Some Famous Theorems

The Four-Colour Theorem

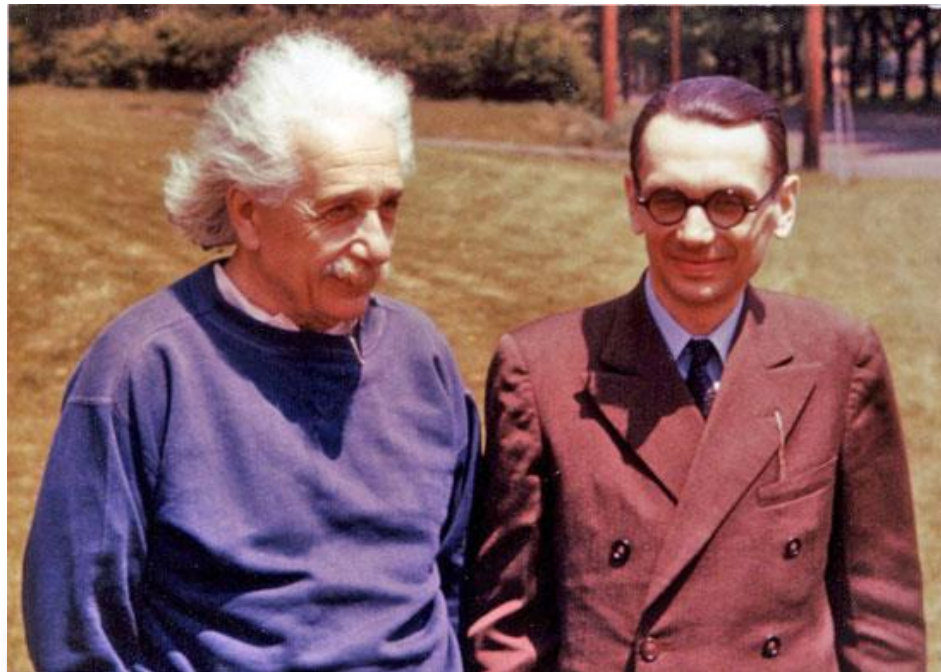
Theorem One needs at most 4 colours to colour any map such that adjacent countries have different colours.



Our inability to prove absolute truths

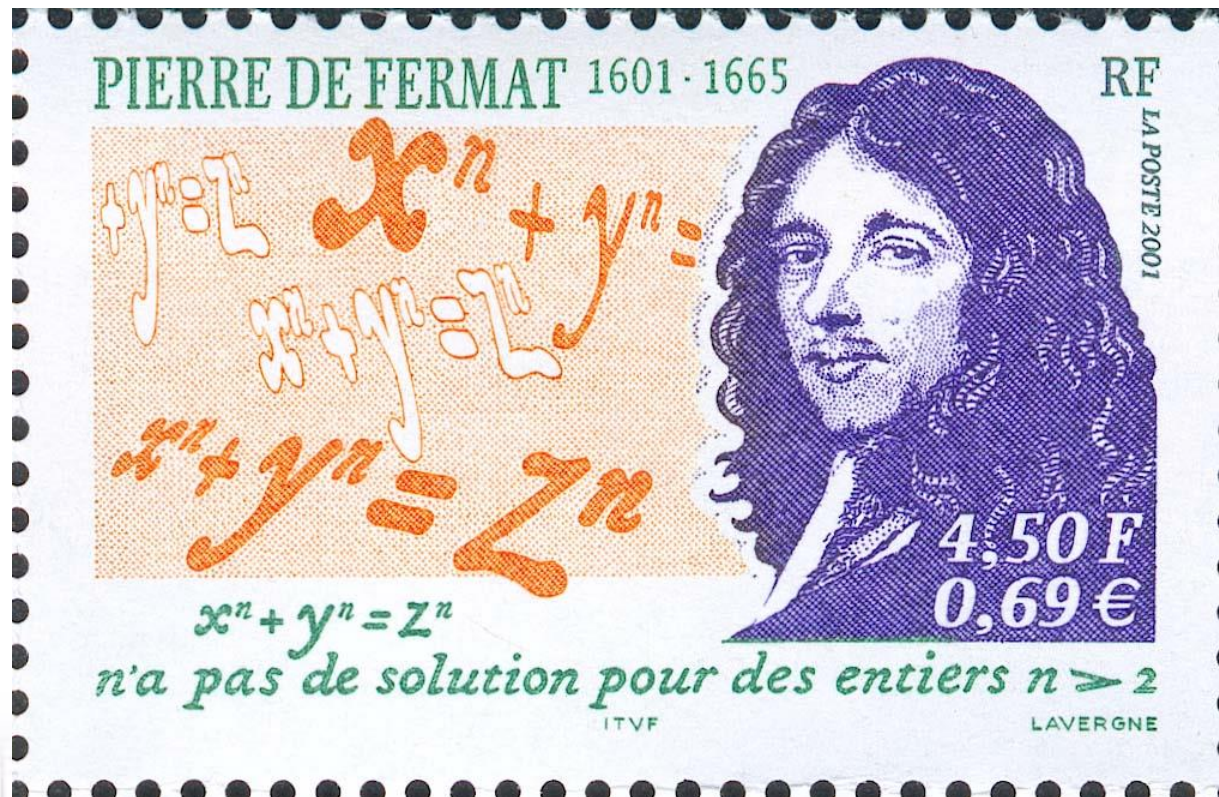
Frege, Russell, and Hilbert's dreams

Gödel's work and the end of a dream



Fermat's Last Theorem

Theorem The equation $x^n + y^n = z^n$, when n is an integer greater than 2, does not have integer solutions.



Other Significant Problems

Poincaré Conjecture (it is a **theorem** now)

Riemann Hypothesis

P versus NP problem

Goldbach's Conjecture